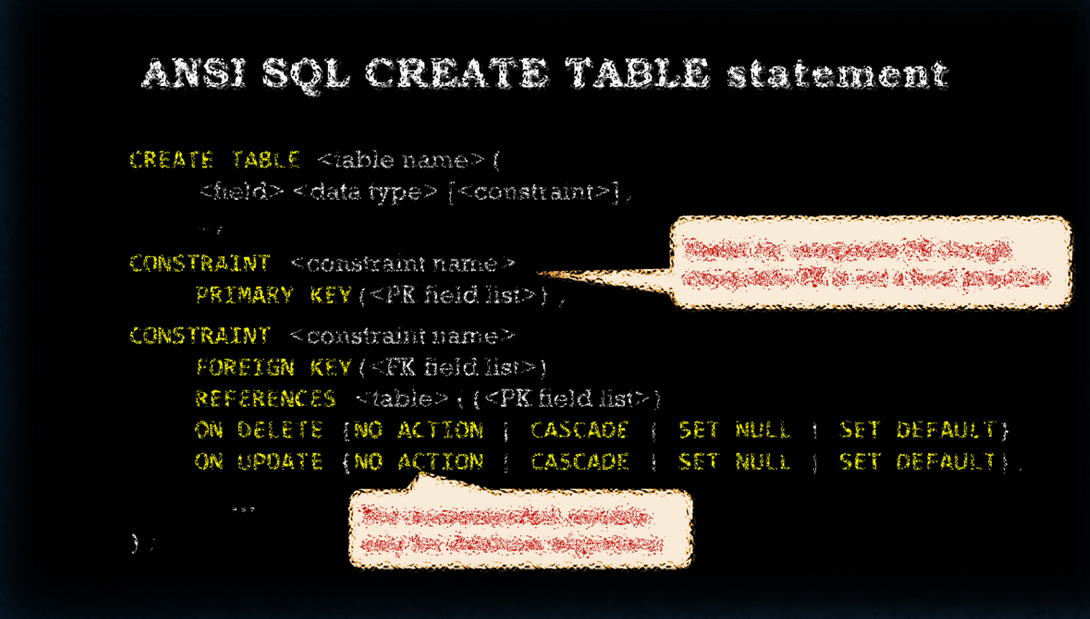
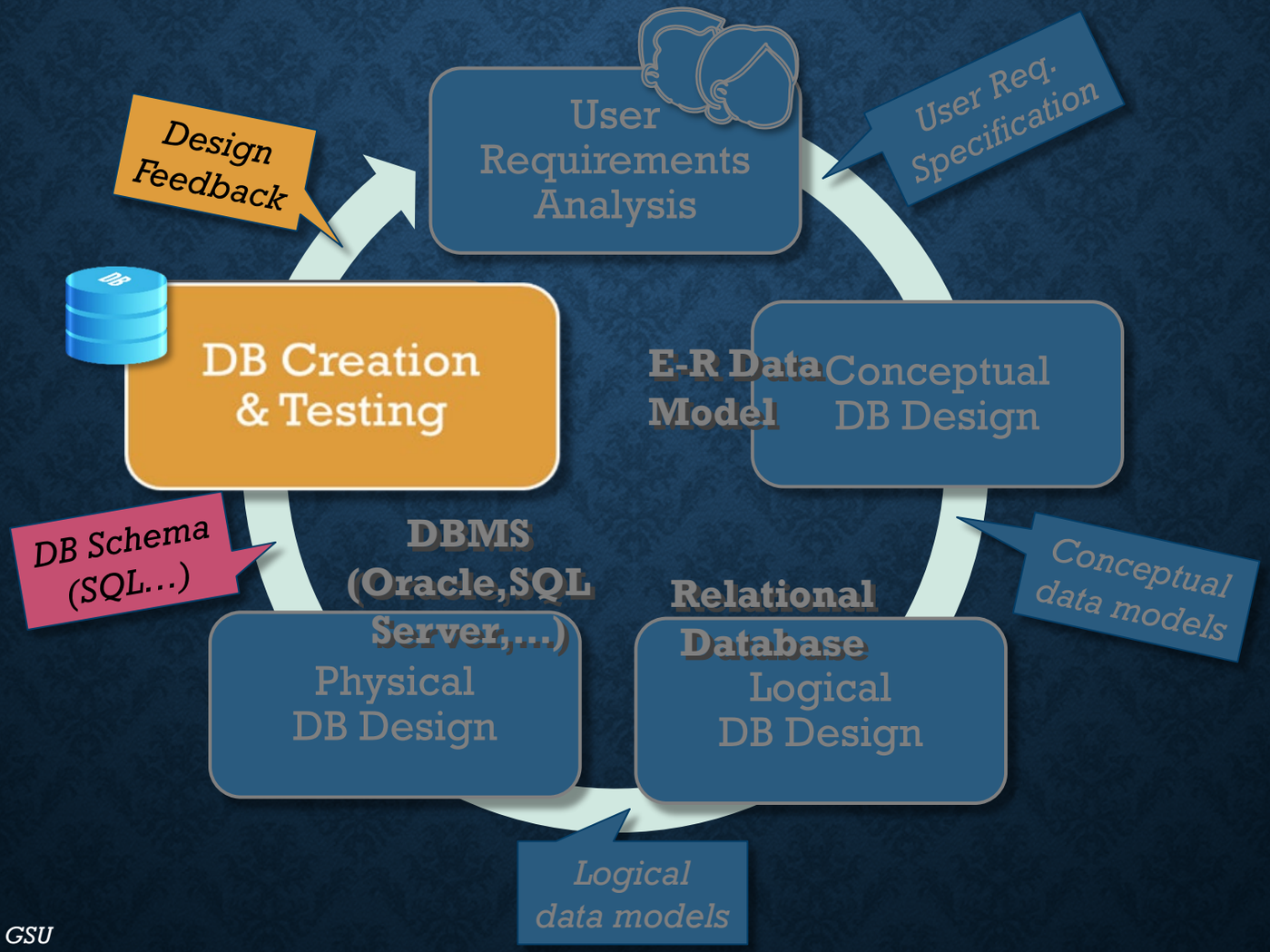
Relational Database & SQL

SQL Table Creation & Changes



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Computer Information Systems, Georgia State University

Revisit the database design lifecycle



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ANSI SQL CREATE TABLE statement

```
CREATE TABLE <table name> (
      <field> <data type> [<constraint>],
      …,
CONSTRAINT <constraint name>
      PRIMARY KEY (<PK field list>),
CONSTRAINT <constraint name>
      FOREIGN KEY (<FK field list>)
      REFERENCES <table> ((<PK field list>)
      ON DELETE {NO ACTION | CASCADE | SET NULL | SET DEFAULT}
      ON UPDATE {NO ACTION | CASCADE | SET NULL | SET DEFAULT},
         …
);
```

For field constraints

Define primary key field(s)

Specify referential integrity constraint(s)

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Explore SQL table creation syntax

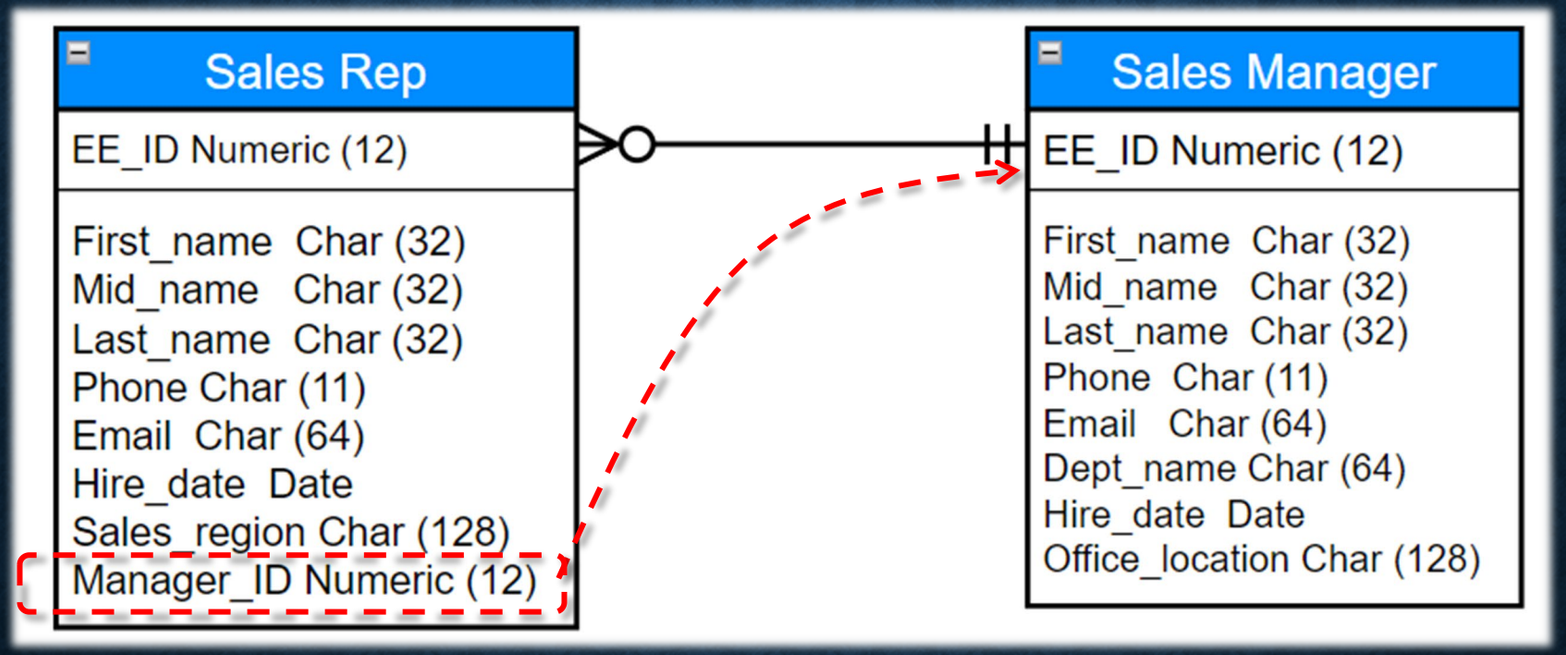
GenAI

Work with your AI tool to

- Explore ANSI SQL field data types and examples
- Understand field data types supported by database vendors
- Study the constraints and explore examples of:
  - Field level constraints
  - Primary key constraints
  - Referential constraint

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Sales Database Schema



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SQL: Create Sales Manager Table

```
CREATE TABLE Sales_Manager (
        EE_ID             numeric(12) PRIMARY KEY,
        first_name        varchar(30) NOT NULL,
        mid_name          varchar(30),
        last_name         varchar(30) NOT NULL,
        phone             varchar(11) NOT NULL,
        email             varchar(64) NOT NULL,
        dept_name         varchar(64),
        hire_date         Date NOT NULL,
        office_location   varchar(128)
);
```

Specify the PK

Specify the required field constraint

The "," separator is required between field definitions

Except for the last definition to avoid a syntax error!

Ending a SQL statement with ";" is a best practice

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SQL: Create Sales Rep Table

```
CREATE TABLE Sales_Rep (
        EE_ID               numeric(12) PRIMARY KEY,
        first_name          varchar(30) NOT NULL,
        mid_name            varchar(30),
        last_name           varchar(30) NOT NULL,
        phone               varchar(11) NOT NULL,
        email               varchar(64) NOT NULL,
        hire_date           Date NOT NULL,
        sales_region        varchar(128),
        manager_ID          numeric (12),
        CONSTRAINT sales_manager_fk FOREIGN KEY (manager_ID)
                   REFERENCES Sales_Manager (EE_ID)
);
```

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SQL: DROP TABLE command

```
DROP TABLE <table>;
```




```
DROP TABLE Sales_Rep;
DROP TABLE Sales_Manager;
```

GenAI

Work with your AI tool to

- Understand the order of table creation and removal
- Discuss if those orders are the same.
- Practice examples of table creation and removal

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SQL: INSERT INTO command

```
INSERT INTO <table> [(<field list>)]
VALUES (<value list>);
```

```
INSERT INTO Sales_Manager (EE_ID, first_name, last_name,
            email, phone, hire_date)
VALUES (210234871951, 'Kathy', 'Milestone',
         'kmilestone@school.edu', 4043219873,
         to_date('05-OCT-2012','DD-MON-YYYY');
```

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SQL: INSERT INTO Table example

```
INSERT INTO Sales_Manager
VALUES (210234871951, 'Kathy', 'LeCat' 'Milestone',
        4043219873, 'kmilestone@school.edu',
        ‘Cosmatic Products',
        to_date('05-OCT-2012','DD-MON-YYYY',
        'Room 420, Building B');
```

- Field names may be omitted if
  - Values are provided for all fields
  - The sequence of values matches that of fields in the CREATE TABLE statement,

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SQL: UPDATE Command

```
UPDATE <table>
   SET      <field> = {<value> | <query>},
            …
[WHERE  <expressions>];
```

- Changes violating integrity constraints would raise errors:
  - Cannot change primary key values to nulls
  - Cannot change the foreign key values referencing to non-existing primary key values

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Update *Sales Manager* table exercises




```
UPDATE Sales_Manager
   SET location = '100 Peachtree St.';

UPDATE Sales_Manager
   SET location = '100 Peachtree St.'
WHERE  first_name = 'Mary';

UPDATE Sales_Manager
   SET location = '100 Peachtree St.'
WHERE  EE_ID = 01301045814;
```

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Collaborate with your AI tool to explain the difference between those UPDATE statements above

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SQL: DELETE Command

```
DELETE FROM <table>
[WHERE <expressions>];
```




```
DELETE FROM Sales_Manager;

DELETE FROM Sales_Manager
WHERE EE_ID = 104585626512;
```

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Collaborate with your AI tool to explain the difference between the DELETE statements above

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SQL DDL and DML commands

- SQL DDL (Data Definition Language) is for creating DB schema
  - Create, Alter, and Drop commands
  - Grant and Revoke privileges and roles
- SQL DML (Data Manipulation Language) is for modifying data
  - Insert, Update, and Delete
  - Other commands

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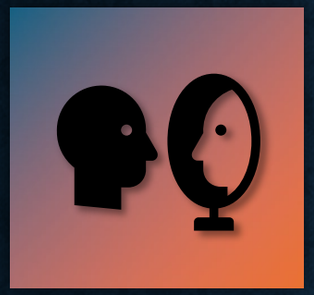
Explore SQL DDL & DML commands

GenAI

Work with your AI tool to

- Explore other ANSI SQL DDL & DDM commands
- Explain the use cases of those commands
- Explore the concept of stored procedures and functions, their use cases, and examples

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Takeaways…

- Covered SQL commands:
  - `CREATE TABLE, DROP TABLE, INSERT INTO, UPDATE, DELETE FROM`
- The sequence of table creation, removal, and changes is very important:
  - A FK table:
    - Can only be created after its referenced PK tables
    - Must be removed prior to remove its referenced PK tables
  - An FK row:
    - Can only be inserted after its referenced PK rows
    - Must be removed prior to remove its reference PK rows
- Common errors are caused by violation of constraints.